Balance equations

(1) Balance of mass

Global statement

$$m := \int_{\mathcal{B}} \rho(\underset{\sim}{x})\, dV = \text{const. in time}$$

ρ: density

The statement $m = \text{const.}$ is equivalent to $\dot{m} = 0$

$$\Rightarrow \quad \dot{m} = \frac{d}{dt}\int_{\mathcal{B}} \rho\, dV = \int_{\mathcal{B}} \dot{\rho}\, dV \overset{!}{=} 0$$

$\rightsquigarrow$ local form:

$$\boxed{\dot{\rho} = 0}$$

(2) Balance of linear momentum

Global statement $\dot{\underset{\sim}{L}} = \underset{\sim}{f}$

with linear momentum $\underset{\sim}{L} := \int_{\mathcal{B}} \rho \dot{\underset{\sim}{x}} \, dv$

and force vector $\underset{\sim}{f} := \underbrace{\int_{\mathcal{B}} \rho \underset{\sim}{\bar{b}} \, dv}_{\text{volume forces}} + \underbrace{\int_{\partial\mathcal{B}} \underset{\sim}{t} \, da}_{\text{boundary forces}}$

$\underset{\sim}{\bar{b}} \downarrow$  $dm = \rho dv$

with $|\underset{\sim}{\bar{b}}| = g$ $\downarrow$ $d\underset{\sim}{\bar{F}} = \underset{\sim}{\bar{b}} dm$

$$\Leftrightarrow \int_{\mathcal{B}} d\underset{\sim}{\bar{F}} = \int_{\mathcal{B}} \rho \underset{\sim}{\bar{b}} \, dv$$

$$\dot{\underset{\sim}{L}} = \frac{d}{dt} \int_{\mathcal{B}} \rho \dot{\underset{\sim}{x}} \, dv = \int_{\mathcal{B}} \dot{\rho \dot{\underset{\sim}{x}}} \, dv$$

$$= \int_{\mathcal{B}} \left(\dot{\rho} \dot{\underset{\sim}{x}} + \rho \ddot{\underset{\sim}{x}} \right) dv$$

$\dot{\rho} = 0$ (balance of mass)

$$= \int_{\mathcal{B}} \rho \ddot{\underset{\sim}{x}} \, dv$$

Insert into global statement:

$$\dot{\underset{\sim}{L}} = \int_{\mathcal{B}} \rho \, \ddot{\underset{\sim}{u}} \, dv = \underset{\sim}{f}$$

special cases

body is homogeneous (ρ = const. in $\mathcal{B}$) and rigid mit $\ddot{\underset{\sim}{u}}$ = const. in $\mathcal{B}$ (e.g. mass point)

$$\rightsquigarrow \int_{\mathcal{B}} \rho \ddot{\underset{\sim}{u}} \, dv = \underset{\sim}{f}$$

$$\Leftrightarrow \rho \ddot{\underset{\sim}{u}} \int_{\mathcal{B}} dv = \rho \ddot{\underset{\sim}{u}} V = \underbrace{m \ddot{\underset{\sim}{u}} = \underset{\sim}{f}}$$

see Engineering Mechanics → "dynamics"

if body moves with constant or zero velocity ($\ddot{\underset{\sim}{u}} = \underset{\sim}{0}$)

$$\rightsquigarrow \int_{\mathcal{B}} \rho \ddot{\underset{\sim}{u}} \, dv = \underbrace{\underset{\sim}{0} = \underset{\sim}{f}} \quad \left(\Rightarrow \sum_i \underset{\sim}{f}_i = \underset{\sim}{0} \right)$$

see Engineering Mechanics → "stereostatics"

General case:

$$\dot{\underset{\sim}{L}} = \int_{\mathcal{B}} \rho \, \ddot{\underset{\sim}{u}} \, dv = \int_{\mathcal{B}} \rho \underset{\sim}{b} \, dv + \int_{\partial\mathcal{B}} \underbrace{\underset{\sim}{t}}_{= \underline{\underline{\sigma}} \underset{\sim}{n}} \, da$$

(Cauchy theorem)

$$\int_{\partial\mathcal{B}} \underline{\underline{\sigma}} \underset{\sim}{n} \, da = \int_{\mathcal{B}} \operatorname{div} \underline{\underline{\sigma}} \, dv$$

(Gauss integral theorem)

$$= \int_{\mathcal{B}} \rho \underset{\sim}{\bar{b}} \, dv + \int_{\mathcal{B}} \operatorname{div} \underset{\sim}{\sigma} \, dv$$

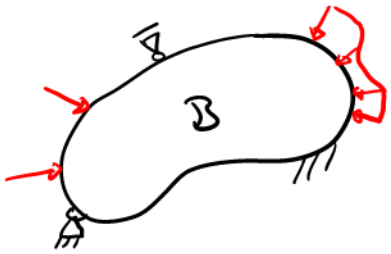
$\rightsquigarrow$ Local form:

$$\boxed{\operatorname{div} \underset{\sim}{\sigma} = \rho (\ddot{\underset{\sim}{u}} - \underset{\sim}{\bar{b}})}$$

if $\ddot{\underset{\sim}{u}} = \underset{\sim}{0}$ and volume forces can be neglected

$\rightsquigarrow \operatorname{div} \underset{\sim}{\sigma} = \underset{\sim}{0}$

Basis for:



Boundary value problem of Mechanics:

Find displacements $\underset{\sim}{u}(\underset{\sim}{x})$ in $\mathcal{B}$, which fulfill

$$\operatorname{div} \underset{\sim}{\sigma} = \rho (\ddot{\underset{\sim}{u}} - \underset{\sim}{b}) \quad \forall \, \underset{\sim}{x} \in \mathcal{B}$$

and the boundary conditions $\underset{\sim}{u} = \bar{\underset{\sim}{u}}$ on $\partial \mathcal{B}_u$

$\underset{\sim}{t} = \bar{\underset{\sim}{t}}$ on $\partial \mathcal{B}_t$

with $\partial \mathcal{B}_u \cap \partial \mathcal{B}_t = \{0\} \wedge \partial \mathcal{B}_u \cup \partial \mathcal{B}_t = \partial \mathcal{B}$

Problem: unknowns are $\underset{\sim}{\sigma}$ and $\underset{\sim}{u}$, but only one vector-valued equation ↯

Solution: constitutive equation $\underset{\sim}{\sigma} := \underset{\sim}{\sigma}(\underset{\sim}{\varepsilon}(\operatorname{Grad} \underset{\sim}{u}))$ provides missing equation to close system of equations

$\Longrightarrow$ material model!

(3) Balance of angular momentum

Global statement: $\dot{\underset{\sim}{h}}_{(O)} = \underset{\sim}{m}_{(O)}$

with angular momentum w.r.t. (O): $\underset{\sim}{h}_{(O)} = \int_{\mathcal{B}} \underset{\sim}{x} \times \rho \dot{\underset{\sim}{x}} \, dv$

and resulting moment $\underset{\sim}{m}_{(O)} = \int_{\mathcal{B}} \underset{\sim}{x} \times \rho \bar{\underset{\sim}{b}} \, dv + \int_{\partial\mathcal{B}} \underset{\sim}{x} \times \underset{\sim}{t} \, da$

(compute time derivative of $\underset{\sim}{h}_{(O)}$, insert and reformulate)

$\vdots$

$\rightsquigarrow$ Local statement

$$\boxed{\underline{\sigma} = \underline{\sigma}^T}$$

④ Balance of energy (1st law of thermodynamics)

Global statement: $\dot{E} + \dot{K} = P + Q$

with internal energy $E = \int_{\mathcal{B}} \rho e \, dv$, $e \hat{=}$ spec. internal energy

kinetic energy $K = \int_{\mathcal{B}} \frac{1}{2} \rho \, \dot{\underset{\sim}{x}} \cdot \dot{\underset{\sim}{x}} \, dv$

power $P = \int_{\mathcal{B}} \rho \, \dot{\underset{\sim}{x}} \cdot \bar{\underset{\sim}{b}} \, dv + \int_{\partial\mathcal{B}} \dot{\underset{\sim}{x}} \cdot \underset{\sim}{t} \, da$

heat $Q = \int_{\mathcal{B}} \rho r \, dv - \int_{\partial\mathcal{B}} \underset{\sim}{q} \cdot \underset{\sim}{n} \, da$

(ρr: heat source, $\underset{\sim}{q}$: heat flow)

Basis for BVP of thermo-mechanics (not focus of this class)

(5) Entropy inequality (2nd law of thermodynamics)

Global statement:

$$\dot{\Gamma} \geq \int_{\mathcal{B}} \frac{s r}{\vartheta}\, dv - \int_{\partial\mathcal{B}} \frac{\underset{\sim}{q}\cdot \underset{\sim}{n}}{\vartheta}\, da$$

ϑ ← absolute temperature

$\Gamma \mathrel{\hat{=}}$ Entropy (thermal energy which can not be directly transformed into mechanical energy)

$$\Gamma := \int_{\mathcal{B}} s\eta\, dv, \quad \eta \mathrel{\hat{=}} \text{spec. entropy (per mass)}$$

Insert balances of mass, linear momentum, and energy yields local statement

$$\vdots$$

$$\boxed{s\left(\vartheta\dot{\eta} - \dot{e}\right) + \underline{\underline{T}}\cdot\underline{\underline{\dot{\varepsilon}}} - \frac{1}{\vartheta}\,\underset{\sim}{q}\cdot \operatorname{grad}\vartheta \geq 0}$$

(Clausius–Duhem–inequality)

Introduce free Helmholtz energy $\tilde{\psi}$ (per mass)

$$\tilde{\psi} := e - \vartheta\eta \rightsquigarrow \dot{\tilde{\psi}} = \dot{e} - \dot{\vartheta}\eta - \vartheta\dot{\eta}$$

$$\Leftrightarrow \dot{e} = \dot{\tilde{\psi}} + \dot{\vartheta}\eta + \vartheta\dot{\eta}$$

$$\Rightarrow (\vartheta\dot{\eta} - \dot{e}) = \cancel{\vartheta\dot{\eta}} - \dot{\tilde{\psi}} - \dot{\vartheta}\eta - \cancel{\vartheta\dot{\eta}} = -(\dot{\tilde{\psi}} + \dot{\vartheta}\eta)$$

$$\Rightarrow \boxed{-\rho(\dot{\tilde{\psi}} + \dot{\vartheta}\eta) + \underline{\underline{\sigma}} \cdot \underline{\underline{\dot{\varepsilon}}} - \frac{1}{\vartheta}\underline{q} \cdot \operatorname{grad}\vartheta \geq 0}$$

(Coleman - Noll inequality)

For isothermal processes ($\dot{\vartheta} = 0$, $\operatorname{grad}\vartheta = \underline{0}$)

$$\rightsquigarrow \boxed{\underline{\underline{\sigma}} \cdot \underline{\underline{\dot{\varepsilon}}} - \rho\dot{\tilde{\psi}} \geq 0}$$

or $\boxed{\underline{\underline{\sigma}} \cdot \underline{\underline{\dot{\varepsilon}}} - \dot{\psi} \geq 0}$ with Helmholtz energy per volume $\psi := \rho\tilde{\psi}$

Basis for construction of thermodynamically consistent material models.